Jonathan Clarke

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EDUCATION

University of Cologne, Cologne, DE
Doctoral Researcher, Advisor: Prof. Dominik Riechers, 2022 – 2027 (anticipated April)
Durham University, Durham, UK
Integrated BSc and MPhys, Physics and Astronomy, 2017 – 2021 (First-Class honours)

AFFILIATIONS

Max Planck Institute for Radio Astronomy, Bonn, DE
Guest Researcher, provided funding, 2022 – present
SFB 1601 CRC, German Science Foundation (DFG),
'Habitats of massive stars across cosmic time", provided funding, 2023 – present
German Astronomical Society (AG), *student affiliation*, 2024 – present
DYNAVERSE Excellence Cluster, *associated member*, 2026 – present

RESEARCH INTERESTS AND EXPERIENCE

Line Intensity Mapping: I investigated statistics of large scale structure across cosmic time. Past work focused on making and analysing wide-field CO, [CII] and [OIII] simulations for FYST/CCAT, TIM, EXCLAIM, SUBLIME-TIFUUN and COMAP, plus simulations for SPHEREx, as well as cross-correlations between them.
High Redshift Galaxies: My work focused on modelling emission of high-z galaxies, e.g. [CII] emission within the ALPINE-ALMA sample near the epoch of reionisation.
Simulations and Modelling: I made light cones using empirical catalogue data, and developed masking pipelines for cleaning line and noise interlopers.
Catalogue Utilisation: I worked with ALPINE-ALMA, COSMOS2020/Web, and Euclid data, including determining their luminosity functions/mass completeness.

PUBLICATIONS

- **Clarke, J.**, Roy, A., et al., Cross-instrument cross-correlation viability for line intensity mapping in the submillimeter regime. In prep, submission expected early 2027 (can discuss upon request)
- **Clarke, J.**, Karoumpis, C., et al., Testing masking effectiveness using multi-line image cubes based on COSMOS 2020 for [CII] line intensity Mapping at $z_{[\mathrm{CII}]} > 3.5$. Arxiv pre-print, in review at A&A
- **Clarke, J.**, Karoumpis, C., et al., [CII] luminosity models and large-scale image cubes based on COSMOS 2020 and ALPINE-ALMA [CII] data back to the epoch of reionisation. 2024, A&A, 689, A1011
- Okada, Y., Schaaf, Reinhold, et al. **(incl. Clarke, J.)**, Observing scheduler for the Fred Young Submillimeter Telescope (FYST). 2026, submitted to A&A
- Saha, H., Honingh, N., **Clarke, J.**, et al., Electromagnetic design and modelling of efficient integrated filterbanks towards a spectroscopic focal plane array for CCAT/FYST. 2026, SPIE Proceedings Volume 14156
- Dev, Ankur., Karoumpis, C., et al. **(incl. Clarke, J.)**, Spectral Data-cube Cleaning for CCAT Deep Spectroscopic Survey. I. Effect of correlated noise and filtering on the power spectrum. 2026, Arxiv pre-print, in review at A&A
- Karoumpis, C., Magnelli, B., et al. **(incl. Clarke, J.)**, [CII] line intensity mapping the epoch of reionization with the Prime-Cam on FYST II. CO foreground masking based on an external catalog. 2024, A&A 691, A262

COLLABORATIONS & OBSERVING PROGRAMS

CCAT Collaboration member, Member of Line Intensity Mapping team
ALMA: Co-I for Cycle 12 submission "Intensity mapping the integrated CII line emission at 400 GHz", code 2025.1.01466.S (PI: Christos Karoumpis)

SELECTED CONFERENCE & WORKSHOP CONTRIBUTIONS

September 2026: Organiser and participant in LIM splinter session for **Astronomische Gesellschaft annual meeting** (Garching)
August 2026: **LIM 2026** (Talk, Pasadena): FYST/CCAT Masking and cross-instrument cross-correlation for submm line intensity mapping
February 2026: **Astroinformatics** (Cologne): Participant in machine learning discussions
June 2025: **LIM 2025** (Talk, Annecy): CCAT: Submillimeter Line Intensity Mapping of [CII] and CO with FYST (on behalf of the CCAT Collaboration)
December 2024: **The multi-color Universe - A new era with Line Intensity Mapping** (Talk, Heidelberg): The viability of cross-correlation predictions for CCAT using existing COSMOS catalogue data
June 2024: **LIM 2024** (Talk, poster, Champaign-Illinois): CCAT, The Deep Spectroscopic Survey with The EoR Spectrometer (on behalf of the CCAT Collaboration)

SELECTED INVITED TALKS

April 2026: **University of Hertfordshire invitation**: Exploring the epoch of reionisation via [CII] line intensity mapping with cross-instrument statistics
October 2025: **Cornell Galaxy Lunch**: Testing masking effectiveness for [CII] line intensity mapping using COSMOS 2020 data
October 2025: **Delft/Cologne Collaboration Meeting**: Overview of preparations for LIM observations with FYST
January 2023: **SFB956 Student Seminar**: Looking at the structure of the early universe with [CII] Luminosity

ADDITIONAL EXPERIENCE

TEACHING:

- Advanced Astrophysics Master's course, University of Cologne (2025 – 2026)
- Scientific Coding Master's course, University of Bonn (via MPIfR) (2025)
- Extension tutor groups, London Academy for Excellence Tottenham (2021–2022)

OUTREACH AND ORGANISATION:

- Volunteer for Long Night of Astronomy Cologne (eclipse, August 2026)
- Astronomy on Tap Köln, organiser, contributor, and speaker (2023 – 2026)
- SFB1601 Student Council Member (2023 – 2025)
- FYST Press Event (April 2024)

CODING & TECHNICAL SKILLS:

- 10 years of Python experience (incl. modelling, simulations, source recovery codes, noise modelling, `astropy`)
- Moderate experience with TOPCAT, GitHub, HTML, LaTeX

OTHER

CITIZENSHIP: UK, USA
LANGUAGES: English (fluent), German (basic)

REFERENCES

Dominik Riechers, *Ph.D. Advisor & Collaborator*: University of Cologne, riechers@ph1.uni-koeln.de
Dongwoo Chung, *Collaborator*: Cornell University, dongwooc@cornell.edu
Frank Bertoldi, *Ph.D. Advisory committee member & Collaborator*: University of Bonn, bertoldi@astro.uni-bonn.de